

Integrable Systems and the Toda Lattice

Antonio Mac Dowell

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Abstract

The goal of this article is to present an introduction to the theory of completely integrable systems, using the celebrated Toda lattice as main example. The Toda lattice is a model for the evolution of n particles on the line with nearest neighbors interactions. The content is divided in two parts. The first part aims to be a general exposition to the theory, in which we give a detailed proof of the fundamental Liouville-Arnold theorem. In the second part, we focus on the open (finite, non-periodic) Toda lattice, showing its integrability and describing action-angle variables. We also show how it can be realized as a flow on a certain coadjoint orbit. The first part of these notes is based on [2, 3, 5] and the second part on [21, 22]. The reader is assumed to have some basic familiarity with symplectic geometry and Lie algebras.

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1 An Overview of integrability

In this section we start by recalling some elementary facts from Poisson manifolds and Hamiltonian systems on them. We also present the concept of bi-Hamiltonian structures along the way and finish with one of the most fundamental theorems about integrable systems, the Liouville-Arnold Theorem.

1.1 Poisson Manifolds

A *Poisson structure* on a manifold M is a *Poisson algebra structure* on its ring of smooth functions. Namely, it is a skew-symmetric bracket $\{\cdot, \cdot\} : C^\infty(M) \times C^\infty(M) \rightarrow C^\infty(M)$ which satisfies the Jacobi identity and the Leibniz rule. Equivalently, this structure has a tensorial description, in terms of a *bivector* $\pi \in \Gamma(\Lambda^2 TM)$, such that

$$[\pi, \pi]_S = 0 \tag{1}$$

where $\Gamma(E)$ denotes the space of sections of a vector bundle E over M and $[\cdot, \cdot]_S$ denotes the *Schouten bracket*: an extension of the Lie bracket for multi-vector fields. The two definitions are related by $\{F, G\} = \pi(dF, dG)$, and equation (1) is equivalent to Jacobi identity of the Poisson bracket. In coordinates, we have

$$\pi(x) = \sum_{i < j} \pi^{ij}(x) \partial_{x_i} \wedge \partial_{x_j}$$

where $\partial_{x_j} = \frac{\partial}{\partial x_j}$ and $\pi^{ij} = \{x_i, x_j\} \in C^\infty(M)$ are the entries of the *Poisson matrix* with respect to these coordinates. Recall that symplectic manifolds have a natural Poisson structure given by $\{F, G\} = \omega(X_G, X_F)$. In this case the *Hamiltonian vector fields* are characterized by the property $\iota_{X_H} \omega = dH$, where ι denotes tensor contraction. Any smooth function on a Poisson manifold $H \in C^\infty(M)$ yields a Hamiltonian vector field $X_H = \{H, \cdot\}$. The equations of motion can be written as

$$\dot{x} := X_H(x(t)) = \sum_{j=1}^n \{H, x_j\} \partial_{x_j}(x(t)) .$$

Functions $F \in C^\infty(M)$ for which $X_F = 0$ are called *Casimirs*. Recall that a bivector yields a *characteristic distribution* in the following way. Consider the map

$$\pi^\flat : T^*M \rightarrow TM , \quad \alpha \mapsto \iota_\alpha \pi .$$

Then $R = \pi^\flat(T^*M) \subset TM$ is a distribution and at each point $x \in M$ we have

$$R_x = \{X_F(x) | F \in C^\infty(M)\} .$$

It turns out that R is integrable, by Weinstein's local splitting theorem [23]. Since Poisson manifolds decompose into symplectic leaves, they are a natural setting for studying families of mechanical systems.

Example 1.1. An important class of examples is that of the canonical Lie-Poisson structure on the dual $\mathfrak{g}^*$ of a (finite dimensional) Lie algebra $\mathfrak{g}$, which originate symplectic leaves known as *coadjoint orbits*. These orbits are obtained from the coadjoint action of a Lie group G on the dual of its Lie algebra. For F, G smooth functions on $\mathfrak{g}^*$, the Poisson bracket is given by

$$\{F, G\}(\xi) = \langle \xi, [d_\xi F, d_\xi G] \rangle \quad (2)$$

where $\langle \cdot, \cdot \rangle$ denotes the pairing a vector space with its dual and the linear functional $dF(\xi) : T_\xi \mathfrak{g}^* \approx \mathfrak{g}^* \rightarrow \mathbb{C}$ is regarded as an element of $\mathfrak{g}$ for any smooth function F on $\mathfrak{g}^*$. Recall that on a coadjoint orbit $\mathcal{O}_\xi$, the Poisson and symplectic structures are related by $\omega_\eta(X_G, X_F) = \{F, G\}(\eta)$, for any $\eta \in \mathcal{O}_\xi$.

If we give $\mathfrak{g}$ a non-degenerate Ad -invariant symmetric bilinear form $\kappa : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ we can use it to define an isomorphism between $\mathfrak{g}$ and $\mathfrak{g}^*$ by contraction. Let us denote such isomorphism and its inverse by $\kappa^\flat$ and ψ , respectively. The Killing form on a semi-simple Lie algebra is an example of such form. This allows us to define a Lie-Poisson structure on $\mathfrak{g}$ as follows.

$$\begin{aligned} \{F, G\}_{\mathfrak{g}}(u) &:= \{\psi^* F, \psi^* G\}(\kappa^\flat(u)) \\ &= \langle \kappa^\flat(u), [d_{\kappa^\flat(u)} \psi^* F, d_{\kappa^\flat(u)} \psi^* G] \rangle \\ &= \kappa(u, [\psi(d_u F), \psi(d_u G)]) . \end{aligned} \quad (3)$$

On the latter equality, observe that since ψ is linear, the chain rule yields

$$d_{\kappa^b(u)}\psi^*F = d_{\kappa^b(u)}(F \circ \psi) = d_uF \circ \psi .$$

Consequently, for any $\xi \in \mathfrak{g}^*$, we have

$$\langle d_{\kappa^b(u)}\psi^*F, \xi \rangle = \langle d_uF, \psi(\xi) \rangle = \kappa(\psi(d_uF), \psi(\xi)) = \langle \xi, \psi(d_uF) \rangle \quad (4)$$

Thus, we can also express our Lie-Poisson bracket on $\mathfrak{g}$ as

$$\{F, G\}_{\mathfrak{g}}(u) = \kappa(u, [\nabla F(u), \nabla G(u)]) , \quad (5)$$

where $\nabla F(u) := \psi(d_uF)$ is the *gradient* of F at $u \in \mathfrak{g}$ with respect to κ . Equivalently, $\nabla F(u)$ is the unique element of $\mathfrak{g}$ such that $\kappa(\nabla F(u), v) = \langle d_uF, v \rangle$ for all $v \in \mathfrak{g}$. Notice that the symplectic leaves of $\mathfrak{g}$ have the same relation with the Poisson bracket as in the dual case, because κ provides an identification between orbits of adjoint and coadjoint orbits. Using (4), we see that the Hamiltonian vector fields on $(\mathfrak{g}, \{\cdot, \cdot\})$ behave like

$$\begin{aligned} X_H(u)[F] &= \langle d_uF, X_H(u) \rangle \\ &= \{H, F\}_{\mathfrak{g}}(u) \\ &= \kappa(u, [\nabla H(u), \nabla F(u)]) \\ &= \kappa([u, \nabla H(u)], \nabla F(u)) \\ &= \langle d_uF, [u, \nabla H(u)] \rangle , \end{aligned}$$

for any function F and any point $u \in \mathfrak{g}$. Therefore, Hamiltonian vector fields on $\mathfrak{g}$ can be expressed as *Lax pairs*:

$$X_H(u) = [u, \nabla H(u)] . \quad (6)$$

The example above motivates the following definition.

Definition 1.1. A *Lax pair* is a pair of time dependent matrices or operators $L(t), P(t)$ acting on a Hilbert space and satisfying

$$\frac{dL}{dt} = [L, P] ,$$

which is known as *Lax's equation*. Here $[L, P] = LP - PL$ is the usual commutator.

More about Lax pairs will be explored further.

1.2 Bi-Hamiltonian Structures

Many Poisson manifolds of interest carry yet another Poisson structure, which is relevant for the Hamiltonian vector fields that we study. We say that two Poisson structures $\{\cdot, \cdot\}_1$ and $\{\cdot, \cdot\}_2$ on a smooth manifold M are *compatible* if any linear combination of them is also Poisson. Then M equipped with these structures is called a *bi-Hamiltonian manifold*.

Let π_1 and π_2 be the bivectors corresponding to $\{\cdot, \cdot\}_1$ and $\{\cdot, \cdot\}_2$, respectively. For non-zero scalars λ_1 and λ_2 we have that $\lambda_1\{\cdot, \cdot\}_1 + \lambda_2\{\cdot, \cdot\}_2$ is a Poisson bracket (associated to the bivector $\lambda_1\pi_1 + \lambda_2\pi_2$) if and only if

$$[\lambda_1\pi_1, \lambda_2\pi_2]_S = 0$$

which is equivalent to $[\pi_1, \pi_2]_S = 0$ since $[\pi_1, \pi_1]_S = [\pi_2, \pi_2]_S = 0$. Therefore, two Poisson structures are compatible if and only if their sum satisfies the Jacobi identity. The following proposition shows how to obtain compatible Poisson structures using Lie derivatives.

Proposition 1.1. *Let $(M, \{\cdot, \cdot\}_1)$ be a Poisson manifold and $X \in \mathfrak{X}(M)$ be any vector field. If $\{\cdot, \cdot\}_2 := \mathcal{L}_X \{\cdot, \cdot\}_1$ satisfies the Jacobi identity, then it is a Poisson structure compatible with $\{\cdot, \cdot\}_1$.*

Proof. Clearly $\{\cdot, \cdot\}_2$ is a Poisson structure. By the remark above, it suffices to show that $\{\cdot, \cdot\} := \{\cdot, \cdot\}_1 + \{\cdot, \cdot\}_2$ satisfies the Jacobi identity. The Lie derivative gives us

$$\{F, G\}_2 = X[\{F, G\}_1] - \{X(F), G\}_1 - \{F, X(G)\}_1 .$$

Then putting all brackets $\{\cdot, \cdot\}_2$ in terms of $\{\cdot, \cdot\}_1$ give the desired equality, in view of the Jacobi identity for $\{\cdot, \cdot\}_1$, applied to 4 cycles. $\square$

We say that a vector field on a bi-Hamiltonian manifold is bi-Hamiltonian when it is Hamiltonian with respect to both Hamiltonian structures.

Example 1.2. Let $F : \mathbb{C}^2 \rightarrow \mathbb{C}$ be a holomorphic function and consider its decomposition into real and imaginary parts $F = G + \sqrt{-1} H$, where G and H are smooth real functions on $\mathbb{R}^4$. The coordinates (z_1, z_2) on $\mathbb{C}^2$ will accordingly decompose as $z_i = x_i + \sqrt{-1} y_i$ for $i = 1, 2$. Let $\Omega = dz_1 \wedge dz_2$ be the standard holomorphic symplectic two-form on $\mathbb{C}^2$. Write $\Omega = \omega_0 + \sqrt{-1} \omega_1$, where ω_0 and ω_1 are real symplectic forms on $\mathbb{R}^4$ given by

$$\omega_0 = dx_1 \wedge dx_2 - dy_1 \wedge dy_2 , \quad \omega_1 = dx_1 \wedge dy_2 - dx_2 \wedge dy_1 .$$

The corresponding Poisson structures $\{\cdot, \cdot\}_0$ and $\{\cdot, \cdot\}_1$ are compatible, since both π_0^{ij} and π_1^{ij} are constant (hence $[\pi_0, \pi_1]_S = 0$). The Cauchy-Riemann equations for F ,

$$\frac{\partial G}{\partial x_i} = \frac{\partial H}{\partial y_i} \quad \text{and} \quad \frac{\partial G}{\partial y_i} = -\frac{\partial H}{\partial x_i} \quad (i = 1, 2)$$

imply that $\{x_1, G\}_0 = \frac{\partial G}{\partial x_2} = \frac{\partial H}{\partial y_2} = \{x_1, H\}_1$, and similarly for the other variables x_2, y_1, y_2 . Thus, $\{\cdot, G\}_0 = \{\cdot, H\}_1$. The same argument shows that $\{\cdot, H\}_0 = -\{\cdot, G\}_1$, and therefore both fields are bi-Hamiltonian.

1.3 Integrable Systems

We say that a function $F \in C^\infty(M)$ on a Poisson manifold is in *involution* with a fixed Hamiltonian $H \in C^\infty(M)$ if $\{F, H\} = 0$. In this case, F is a *constant of motion* or *first integral* of H . Notice that F is preserved by the flow of X_H . In general, a subset of $C^\infty(M)$ is called *involutive* if any two elements of it are in involution. Observe that the subalgebra of $C^\infty(M)$ generated by an involutive subset is also involutive.

Example 1.3. Consider a bi-Hamiltonian vector field $X \in \mathfrak{X}(M)$ on a bi-Hamiltonian manifold $(M, \{\cdot, \cdot\}_1, \{\cdot, \cdot\}_2)$. The functions F and G for which $X = \{\cdot, F\}_1 = \{\cdot, G\}_2$ are in involution with respect to both brackets. Indeed, $\{F, G\}_2 = \{F, F\}_1 = 0$ and $\{G, F\}_1 = \{G, G\}_2 = 0$. More generally, suppose we have a *bi-Hamiltonian hierarchy*: a sequence of functions $(F_i)_{i \in \mathbb{Z}} \subset C^\infty(M)$ such that

$$\{\cdot, F_i\}_2 = \{\cdot, F_{i+1}\}_1 , \quad i \in \mathbb{Z} .$$

Then for $i < j$ one has

$$\{F_i, F_j\}_1 = \{F_i, F_{j-1}\}_2 = \{F_{i+1}, F_{j-1}\}_1 = \dots = \{F_j, F_i\}_1 ,$$

and therefore $\{F_i, F_j\}_1 = 0$ by skew-symmetry. Thus, $(F_i)_{i \in \mathbb{Z}} \subset C^\infty(M)$ is involutive with respect to both Poisson structures.

Example 1.4. Let $W \subset \mathfrak{gl}(n)$ be an affine subspace of the Lie algebra of $n \times n$ matrices and let H be a Hamiltonian on W . As seen in example 1.1, we may suppose that there exists a Poisson structure on W such that the Hamiltonian vector field X_H is a Lax pair

$$\dot{M} := X_H(M) = [M, Y] \in \mathfrak{gl}(n) ,$$

where $M \in W$ and $Y : W \rightarrow \mathfrak{gl}(n)$ is such that $[M, Y(M)] \in W$ for all M . We claim that all coefficients of the characteristic polynomial $\det(\lambda Id - M)$ are in involution with H .

To see this, let $M(t)$ be an integral curve of $\dot{M}$, defined for $|t| < \epsilon$, with $\epsilon > 0$ sufficiently small, and denote by $Y(t)$ the corresponding value of Y at $M(t) \in W$. For any positive integer k , one computes

$$\begin{aligned} \{H, Tr(M^k)\} &= \frac{d}{dt} Tr(M^k(t)) \\ &= k Tr \left(M^{k-1}(t) \frac{dM}{dt}(t) \right) \\ &= k Tr (M^k(t)Y(t) - M^{k-1}(t)Y(t)M(t)) = 0 . \end{aligned}$$

Thus, the functions $H_k : M \mapsto Tr(M^k)$ are in involution with H . The same is true for each coefficient of $|\lambda Id - M|$, for they are polynomial functions of the H_k 's. This implies the *isospectral* property of Lax pairs: eigenvalues are preserved.

We now give a precise definition of integrability.

Definition 1.2. Let (M^{2n}, ω) be a $2n$ -dimensional symplectic manifold. We say that a Hamiltonian H on M is *Liouville integrable* (or just *integrable*) if and only if there exists n functions $F_1 = H, \dots, F_n$ such that

1. $\{F_i, F_j\} = 0$ for all $1 \leq i, j \leq n$;
2. the differentials $dF_1, \dots, dF_n$ are linearly independent almost everywhere on M ;
3. the Hamiltonian vector fields X_{F_i} are complete (i.e. their flows are defined for all $t \in \mathbb{R}$).

There is a similar and more general notion of integrable systems on Poisson manifolds but we shall omit that, for the sake of simplicity. The interested reader may consult [2].

The common level sets of $F_1, \dots, F_n$ give a foliation on M , called *Liouville foliation* (with respect to H). Since the F_i 's are preserved along the flow of X_H , each leaf is an invariant submanifold. There are *regular leaves* (corresponding to regular values of $\mathbf{F} := (F_1, \dots, F_n)$) filling almost all M , and *singular leaves*, filling the remaining zero measure set. As we shall see, the *Liouville-Arnold theorem* describes the Liouville foliation around the regular leaves.

Remark 1.1. An immediate consequence of the Jacobi identity one obtains the classical result, known as *Poisson Theorem*: If $\{F, H\} = \{G, H\} = 0$, then $\{\{F, G\}, H\} = 0$. Another classical theorem that leads to constants of motion is *Noether's Theorem*. Recall that a Hamiltonian action of a Lie group on a Poisson manifold yields a (co-)moment map [3]. Let us denote by $\chi : G \times M \rightarrow M$ such action, with $\chi_g : M \rightarrow M$ being the diffeomorphism of M induced by $g \in G$. We also denote the infinitesimal generators of the action by $v_M(p) = \frac{d}{dt}|_{t=0} \chi_{\exp(tv)}(p)$ for each $v \in \mathfrak{g}$ and $p \in M$.

Theorem 1.1 (Noether). *Let G be a Lie group acting Hamiltonianly on a Poisson manifold $(M, \{\cdot, \cdot\})$, with co-moment map $\mu : \mathfrak{g} \rightarrow C^\infty(M)$. If H is a G -invariant function ($H = H \circ \chi_g \forall g \in G$) then for any $v \in \mathfrak{g}$, the function $\mu(v)$ is a constant of motion for H .*

Proof. Let $p \in M$ be arbitrary. By G -invariance of H we have that

$$-X_H(p)[\mu(v)] = \{H, \mu(v)\}(p) = v_M(p)[H] = \frac{d}{dt} \Big|_{t=0} H(\chi_{\exp(tv)}(p)) = \frac{d}{dt} \Big|_{t=0} H(p) = 0 .$$

□

The point here is to emphasize that Hamiltonian vector fields generated by group actions lead to Lie algebra induced constants of motion.

Example 1.5. Let $M = \mathbb{R}^{2n}$ with the standard symplectic form $\omega = \sum_{i=1}^n dq_i \wedge dp_i$ and Hamiltonian

$$H = \frac{1}{2} \sum_{i=1}^n p_i^2 + \nu_i q_i^2 , \quad \nu_i > 0 . \quad (7)$$

This is known as the *harmonic oscillator* between physicists. It is easy to check that the functions $F_i := \frac{1}{2}(p_i^2 + \nu_i q_i^2)$ are independent and in involution and thus $\mathbf{F}$ is completely integrable. Notice that the fibers $\mathbf{F}^{-1}(c)$ for $c = (c_1, \dots, c_n) \in \mathbb{R}^n$ are products of the circles $\{p_i^2 + \nu_i q_i^2 = c_i\}$ and so they are n -dimensional tori. Also observe that the functions $q_i p_j - q_j p_i$ are all constants of motion, but they are not in involution with each other.

Let $M_c := \mathbf{F}^{-1}(c)$ for a fixed $c = (c_1, \dots, c_n) \in \mathbb{R}^n$ denote a regular leaf from the Poisson foliation of an integrable system on (M^{2n}, ω) . Now we are more then ready for the main theorem. We follow [3, 5].

Theorem 1.2 (Liouville-Arnold). *Let (M^{2n}, ω, H) be a real integrable system with $\mathbf{F} = (F_1 = H, \dots, F_n)$, and let M_c be as above. Then*

1. M_c is a Lagrangian submanifold, invariant under the flow of the Hamiltonian vector fields X_{F_i} .
2. If M_c is compact and connected, then it is diffeomorphic to the n -dimensional torus $\mathbb{T}^n = (\mathbb{R}/\mathbb{Z})^n$. In this case, we call M_c a Liouville torus.
3. the Liouville foliation is trivial on some neighborhood of the Liouville torus M_c . I.e., there exists some neighborhood $\mathcal{U}$ of M_c such that $\mathcal{U} \cong \mathbb{T}^n \times \mathbb{D}^n$, where $\mathbb{D}^n$ is the n -disk in $\mathbb{R}^n$.
4. On $\mathcal{U} \cong \mathbb{T}^n \times \mathbb{D}^n$ there are coordinates $(s_1, \dots, s_n, \theta_1, \dots, \theta_n)$ (called action-angle coordinates), such that

$$(a) \quad \omega = \sum d\theta_i \wedge ds_i .$$

$$(b) \quad \text{The action coordinates } s_i \text{ are functions of } F_1, \dots, F_n ;$$

$$(c) \quad \text{In the coordinates } (s, \theta), \text{ the flow of } H = F_1 \text{ is straightened on each of the Liouville torus in the neighborhood } \mathcal{U}. \text{ Strictly speaking, we have}$$

$$\dot{s}_i = 0 , \quad \dot{\theta}_i = q_i(s_1, \dots, s_n) .$$

This means that the flow of H determines a conditionally periodic motion that generates a rational or irrational linear winding on each torus.

Remark 1.2. *In view of the first claim of theorem 1.2, integrable systems can be regarded as Lagrangian fibrations.*

Proof. (1): Of course $M_c \subset M$ is a n -submanifold, by the inverse function theorem. Since $\{F_i, F_j\} = 0$, we have that F_i is preserved along the flow of X_{F_j} . Therefore, $M_c = \mathbf{F}^{-1}(c)$ is invariant under all these flows. Moreover, the vector fields X_{F_i} are linearly independent and span the whole tangent space of M_c . This is true because the contraction isomorphism $\omega^\flat : \mathfrak{X}(M_c) \rightarrow \Omega^1(M_c)$ sends the 1-forms dF_i to X_{F_i} . Thus, since $\omega(X_{F_i}, X_{F_j}) = \{F_j, F_i\} = 0$, we have that $\omega|_{M_c} \equiv 0$. It follows that M_c is a Lagrangian invariant submanifold.

(2): Since $[X_{F_i}, X_{F_j}] = X_{\{F_i, F_j\}} = 0$, the flows of X_{F_i} and X_{F_j} commute and are complete, by hypothesis. This allows us to define an action of the abelian group $\mathbb{R}^n$ on M^{2n} as follows.

Let ψ_i^t denote the flow of X_{F_i} . Then, for $(t_1, \dots, t_n) \in \mathbb{R}^n$, define

$$\Psi(t_1, \dots, t_n) := \psi_1^{t_1} \circ \dots \circ \psi_n^{t_n} \in \text{Diff}(M)$$

Lemma 1.1. *If the submanifold M_c is connected, then it is an orbit of the $\mathbb{R}^n$ -action defined above.*

Proof. Let $x \in M_c$ and consider the map $A_x(t_1, \dots, t_n) := \Psi(t_1, \dots, t_n)(x)$. Notice that it is a local diffeomorphism onto its image. Indeed, denoting the canonical vector fields on $\mathbb{R}^n$ by e_i , we have

$$dA_x(e_i) = \left. \frac{d}{dt} \right|_{t=0} A_x(te_i) = \left. \frac{d}{dt} \right|_{t=0} \Psi(e_i)(x) = \left. \frac{d}{dt} \right|_{t=0} \psi_i^t(x) = X_{F_i}(x) .$$

Since the X_{F_i} are linearly independent and M_c is n dimensional, it follows that A_x is a local diffeomorphism.

Thus, $\mathcal{O}_x = A_x(\mathbb{R}^n)$ is an open set in M_c . Now, suppose that $M_c \not\subset \mathcal{O}_x$. Then it must be at least the union of two orbits $\mathcal{O}_x$ and $\mathcal{O}_y$, for $y \in M_c$. But $\mathcal{O}_y$ is also open, which implies that M_c is a disjoint union. But this violates the connectivity of M_c . □

Lemma 1.2. *An orbit of maximal dimension $\mathcal{O}_x$ of the $\mathbb{R}^n$ -action Ψ is the quotient $\mathbb{R}^n / \mathbb{Z}^k$. In particular, if $\mathcal{O}_x$ is compact, then $k = n$ and hence $\mathcal{O}_x \cong \mathbb{T}^n$.*

Proof. Every orbit $\mathcal{O}_x$ of the smooth action Ψ is the quotient of $\mathbb{R}^n$ by the stabilizer of x , denoted by S_x . Since A_x is a local diffeomorphism, S_x is a discrete subgroup of $\mathbb{R}^n$ (and therefore has no accumulation points). In particular, inside any bounded set of $\mathbb{R}^n$ there is only a finite number of points in S_x .

We claim that S_x is a lattice: $S_x \cong \mathbb{Z}^k$. I.e., all points of $\mathbb{R}^n$ that leave x still are obtained as linear combinations with integer coefficients of k generators. We proceed by induction.

Suppose $n = 1$. Let $e_1 \neq 0$ be the element of $S_x \subset \mathbb{R}$ that is closest to zero. If there is some $y \in S_x$ that cannot be written as an integral multiple of e_1 , then for some $k \in \mathbb{Z}$ we have

$$ke_1 < y < (k+1)e_1 .$$

But then $y - ke_1$ is closer to zero than e_1 , contradiction. Thus, $S_x = \{ke_1 | k \in \mathbb{Z}\}$.

Suppose $n = 2$. Let $0 \neq e_1 \in S_x \subset \mathbb{R}^2$ be the closest to the origin and consider the line generated by it, $\mathbb{R}e_1$. There are two possible scenarios:

- All the other elements $y \in S_x$ lie on the line $\mathbb{R}e_1$. But this implies our claim.
- Otherwise, there exists some $y \in S_x$ such that $y \notin \mathbb{R}e_1$. In this case, take $e_2 \neq 0$ as the element of S_x that is closest to $\mathbb{R}e_1$ (and also closest to the origin), but does not belong to it. We will see that any $y \in S_x$ can be written as $k_1e_1 + k_2e_2$, for $k_1, k_2 \in \mathbb{Z}$. Suppose that there is some $h \in S_x$ that does not admit such expression. Consider the grid on $\mathbb{R}^2$ generated by the integer combinations of e_1 and e_2 (see figure 1). Notice that h must lie inside one of the parallelograms of the grid (otherwise it falls in one of the cases already treated). We can translate h by an appropriate integer combination of e_1 and e_2 to obtain h' closer to $\mathbb{R}e_1$ than e_2 , which is a contradiction.

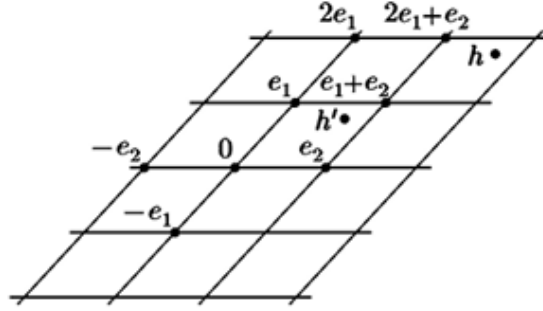


Figure 1: The grid on $\mathbb{R}^2$. Stolen from [5].

Thus, $S_x = \{k_1e_1 + k_2e_2 | k_1, k_2 \in \mathbb{Z}\}$. Continuing this argument by induction, we obtain that $S_x \subset \mathbb{R}^n$ is a lattice generated by $e_1, \dots, e_k$, for some $k \leq n$.

If $k < n$, then the quotient space $\mathbb{R}^n / \mathbb{Z}^k$ is a cylinder $\mathbb{T}^k \times \mathbb{R}^{n-k}$. In particular, if the orbit is compact, then $n = k$ and $\mathcal{O}_x \cong \mathbb{T}^n$.

□

(3): The existence of a neighborhood $\mathcal{U}$ of M_c which is diffeomorphic to the product $\mathbb{T}^n \times \mathbb{D}^n$ comes from a well known and fundamental theorem, that we state as follows.

Let $F : M \rightarrow N$ be a smooth map between smooth manifolds, and let $y \in N$ be a regular value of F . Suppose that $F^{-1}(y)$ is compact. Then there exists a neighborhood $\mathcal{V} \subset N$ such that $F^{-1}(\mathcal{V}) \cong \mathcal{V} \times F^{-1}(y)$. Moreover, the restriction $F|_{\mathcal{V} \times F^{-1}(y)} : \mathcal{V} \times F^{-1}(y) \rightarrow \mathcal{V}$ is just the projection onto the first factor, and in particular, for every $z \in \mathcal{V}$, we have $F^{-1}(z) \cong F^{-1}(y)$.

(4): We proceed to the construction of the action-angle variables. Consider a neighborhood of a Liouville torus, $\mathcal{U}(M_c) \cong \mathbb{T}^n \times \mathbb{D}^n$. Choose a point x on each torus $T = \mathbb{R}^n / S_x$, varying smoothly, so that the basis $e_1, \dots, e_n$ of S_x also varies smoothly with x . This can be done because the coordinates $e_i = (t_1, \dots, t_n) \in \mathbb{R}^n$ are the solutions of $\Psi(t_1, \dots, t_n)(x) = x$ (by the definition of S_x), where x is a parameter. By the inverse function theorem, the solutions depend smoothly on these parameters. Notice that we may apply the inverse function theorem here because $\frac{\partial \Psi(t)}{\partial t_j}(x) = X_{F_i}(\Psi(t)(x))$ are linearly independent.

Define charts $(\phi_1, \dots, \phi_n)$ on the torus M_c in the following manner: for each $y = \Psi(a)(x) \in M_c$, where $a = a_1e_1 + \dots + a_ne_n \in \mathbb{R}^n$, set

$$\phi_1(y) = 2\pi a_1(\text{mod } 2\pi), \dots, \phi_n(y) = 2\pi a_n(\text{mod } 2\pi). \quad (8)$$

We claim that the coordinates $(\phi_1, \dots, \phi_n)$ are such that the vector fields ∂_{ϕ_i} and X_{F_i} are related by the following linear change of basis with constant coefficients

$$\partial_{\phi_i} = \sum_k c_{ik} X_{F_i} . \quad (9)$$

Indeed, for the coordinates $(F_1, \dots, F_n, \phi_1, \dots, \phi_n)$ on $\mathcal{U} = \mathcal{U}(M_c)$, we have

$$\omega = \sum_{ij} \tilde{c}_{ij} dF_i \wedge d\phi_j + \sum_{ij} b_{ij} dF_i \wedge dF_j . \quad (10)$$

Then contraction with ∂_{ϕ_i} yields the desired relation (9). Notice that there is no term of the form $a_{ij} d\phi_i \wedge d\phi_j$, since all Liouville tori on $\mathcal{U}$ are Lagrangian.

Now we claim that $\tilde{c}_{ij} = c_{ij}$ and in particular do not depend on $\phi_1, \dots, \phi_n$. Indeed,

$$\begin{aligned} \tilde{c}_{ij} &= \omega(\partial_{F_i}, \partial_{\phi_j}) = \omega(\partial_{F_i}, \sum_k c_{kj} X_{F_k}) \\ &= \sum_k c_{kj} \omega(\partial_{F_i}, X_{F_k}) = \sum_k c_{kj} dF_k(\partial_{F_i}) \\ &= \sum_k c_{kj} \frac{\partial F_k}{\partial F_i} = c_{ij} = c_{ij}(F_1, \dots, F_n) . \end{aligned}$$

We also claim that the coefficients b_{ij} also do not depend on the coordinates ϕ_i . Since ω is closed, using expression (10) and that $c_{ij} = -c_{ji}$ and $b_{ij} = -b_{ji}$ we get

$$\frac{\partial b_{ij}}{\partial \phi_k} = \frac{\partial c_{kj}}{\partial F_i} - \frac{\partial c_{ki}}{\partial F_j} ,$$

so $\frac{\partial b_{ij}}{\partial \phi_k}$ does not depend on ϕ_k . But b_{ij} is a 2π -periodic function of ϕ_k on the torus, so our claim is verified.

The claims above give an important corollary. Write $\omega = \sum_i \omega_i \wedge d\phi_i + \beta$, where

$$\omega_i = \sum_j c_{ij} dF_j \in \Omega^1(\mathbb{D}^n) , \quad \beta = \sum_{ij} b_{ij} dF_i \wedge dF_j \in \Omega^2(\mathbb{D}^n)$$

are closed (because ω is closed) and do not depend on $\phi_1, \dots, \phi_n$.

Lemma 1.3. *On the neighborhood $\mathcal{U} = \mathcal{U}(M_c)$ there exists $\alpha \in \Omega^1(\mathcal{U})$ such that $\omega = d\alpha$. In other words, ω is exact.*

Proof. It follows from a result known as *homotopy invariance*: Let $Y \subset X$ be a submanifold admitting a map $X \rightarrow Y \subset X$ that is homotopic to the identity on X . Then a closed differential form η on X is also exact if and only if its restriction to Y , $\eta|_Y$ is also exact.

Alternatively, we can prove the lemma by straightforward calculation: Since ω_i and β are closed differential forms on the disk $\mathbb{D}^n$, by Poincaré lemma they are also exact: $ds_i = \omega_i$, $d\xi = \beta$. Take $\alpha = \sum s_i d\phi_i + \xi$. Then

$$d\alpha = \sum ds_i \wedge d\phi_i + d\xi = \sum \omega_i \wedge d\phi_i + \beta = \omega .$$

□

Consider now the functions $s_1 = s_1(F_1, \dots, F_n), \dots, s_n = s_n(F_1, \dots, F_n)$ such that $ds_i = \omega_i$. By the definition of ω_i it is clear that $c_{ij} = \frac{\partial s_i}{\partial F_j}$. Let Ω be the matrix of the symplectic structure ω , which has the form

$$\Omega = \begin{pmatrix} 0 & c_{ij} \\ -c_{ij} & b_{ij} \end{pmatrix}.$$

It follows that $0 \neq \det(\Omega) = (\det(C))^2$, where $C = (c_{ij})$ is the Jacobian matrix of the transformation $s_1 = s_1(F_1, \dots, F_n), \dots, s_n = s_n(F_1, \dots, F_n)$. Thus, we can take $(s_1, \dots, s_n, \phi_1, \dots, \phi_n)$ as a system of coordinates on $\mathcal{U}$.

Now write $\xi = \sum g_i \wedge ds_i$ and perform the following change of variables

$$\theta_i = \phi_i - g_i(s_1, \dots, s_n).$$

Geometrically, this means a change on the initial reference points for the angle coordinates on the Liouville tori. Notice that the level sets and the generators chosen in the beginning of the proof remain unaltered. Finally, we compute

$$\begin{aligned} \sum ds_i \wedge d\theta_i &= \sum ds_i \wedge d(\phi_i - g_i(s_1, \dots, s_n)) \\ &= \sum ds_i \wedge d\phi_i + \sum dg_i(s_1, \dots, s_n) \wedge ds_i \\ &= \sum ds_i \wedge d\phi_i + d\xi \\ &= \sum ds_i \wedge d\phi_i + \beta = \omega, \end{aligned}$$

and so the action-angle variables are constructed.

The last thing that remains to be proven is that the flow of X_H is straightened on the Liouville tori with the coordinates (s, θ) . Indeed, we have that $X_{s_i} = \partial_{\theta_i}$ and consequently $\frac{\partial H}{\partial \theta_i} = X_{s_i}(H) = \{H, s_i\} = 0$. Thus, $H = H(s_1, \dots, s_n)$ is constant on the Liouville tori and therefore, $X_H = \sum \frac{\partial H}{\partial s_i} X_{s_i} = \sum \frac{\partial H}{\partial s_i} \partial_{\theta_i}$, implying that

$$\frac{\partial H}{\partial s_i} = \frac{\partial H}{\partial s_i}(s_1, \dots, s_n),$$

which are constant on the Liouville tori. □

Remark 1.3. *The action variables can be defined explicitly. Let $\mathcal{U}(M_c) \cong \mathbb{D}^n \times \mathbb{T}^n$ and fix a basis $e_1, \dots, e_n$ on the lattice associated to M_c . This determines (uniquely) a basis $\gamma_1, \dots, \gamma_n$ of $\pi_1(\mathbb{T}^n) \cong \mathbb{Z}^n$. By continuity, these cycles extend to all Liouville tori on $\mathcal{U}$. For each tori, we associate the numbers $s_1, \dots, s_n$ given by*

$$s_i = \frac{1}{2\pi} \oint_{\gamma_i} \alpha,$$

where $\alpha \in \Omega^1(\mathcal{U})$ is such that $d\alpha = \omega$ and is often called the action form. This gives us n smooth functions on $\mathcal{U}$, $s_1 = s_1(F_1, \dots, F_n), \dots, s_n = s_n(F_1, \dots, F_n)$ that coincide (up to a constant) with the action variables, as we shall see next.

Let L_c be a family of smooth Lagrangian submanifolds on (M^{2n}, ω) , where c is a parameter with values on a simply connected region C (say, a disk $\mathbb{D}^n \subset \mathbb{R}^k$). In our case, these are all Liouville tori parametrized by values of the first integrals F_i .

Let $[\gamma] \in \pi_1(L_c)$. Since the parameter of this family varies on a simply connected region, we may assume that $[\gamma]$ is naturally fixed in the fundamental group of each Lagrangian submanifold L_c , for $c \in C$.

If $\omega = d\alpha$, then for each L_c we associate the real number $s(c)$ (called the action related to the fixed element $[\gamma] \in \pi_1(L_c)$) via

$$s_\gamma(c) = \frac{1}{2\pi} \oint_{\gamma_c} \alpha ,$$

where $\gamma_c \subset L_c$ is a representative of the fixed class $[\gamma]$. Thus, we have a function (associated to $[\gamma]$), denoted by $s : C \rightarrow \mathbb{R}$, which we call *action*. If ω is not exact, we can make an analogous construction (see [5]).

To end this section we outline a few general properties of the actions $s_i(c)$.

1. *The value of $s_\gamma(c)$ does not depend on the choice of the curve in the homotopy class of γ .* Indeed, the 1-form α restricted to M_c is closed, so using Stokes' theorem we obtain equality of integrals over homologous cycles.
2. *The action s_i is defined up to a constant.* Notice that there is an ambiguity in the choice of α : if $\alpha' = \alpha + \beta$, for some $d\beta = 0$ (so that $d\alpha = d\alpha' = \omega$), then

$$s'_\gamma(c) = \frac{1}{2\pi} \oint_{\gamma_c} \alpha' = \frac{1}{2\pi} \left(\oint_{\gamma_c} \alpha + \oint_{\gamma_c} \beta \right) = s_\gamma(c) + \frac{1}{2\pi} \oint_{\gamma_c} \beta .$$

Thus, s_γ and s'_γ differ by a constant that depends on β but does not depend on the chosen Liouville torus (or Lagrangian submanifold).

3. *Let $\mathcal{U}(M_c) \cong \mathbb{D}^n \times \mathbb{T}^n$. Fix a nontrivial cycle γ on each torus of $\mathcal{U}$, varying smoothly. Consider the corresponding action*

$$s_\gamma(c) = \frac{1}{2\pi} \oint_{\gamma_c} \alpha .$$

Then all the trajectories of the Hamiltonian vector fields X_{s_γ} are closed, with period 2π and are homologous to γ . Indeed, we can take $\gamma = \gamma_1$, corresponding to θ_1 in $(s_1, \dots, s_n, \theta_1, \dots, \theta_n)$, without loss of generality. In particular, $s_\gamma = s_1$, so $X_{s_\gamma} = \partial_{\theta_1}$, and our statement becomes obvious.

2 The Toda Lattice

The Toda lattice is a system of particles on the line with exponential interaction of nearest neighbors. Here we restrict our attention to the case of finitely many particles, without periodicity. For the periodic Toda lattice, where the first and last particles interact with each other as neighbors, the interested reader can consult [20].

We now give a few words on the history of the subject, following [21, 22]. In the mid sixties, the Japanese physicist M. Toda [14] proposed a model for wave propagation along n particles in a line with the Hamiltonian

$$H(q, p) = \frac{1}{2} \sum_{k=1}^n p_k^2 + \sum_{k=1}^{n-1} \exp(q_k - q_{k+1}) , \quad (11)$$

where q and p represent the positions and momenta of these particles at a given time, respectively. We consider the nonperiodic case by taking $q_0 = -\infty$ and $q_{n+1} = +\infty$. It is worth mentioning that here the particles do not collide, they simply pass each other.

In 1974 Flaschka and Manakov [7, 15] showed that the system is completely integrable. Soon after that Moser Kac and van Moerbeke [9, 17] showed that for the nonperiodic lattice, the quantities $\exp(q_j(t))$ are rational functions of $\exp(\lambda_k t)$ where λ_k is the asymptotic momentum of the k -th particle as $t \rightarrow +\infty$. In 1976, Bogoyavlensky [4] introduced the generalized Toda lattices associated to root systems of simple Lie algebras. The original Toda lattice is related to the Lie algebra $\mathfrak{sl}(n, \mathbb{R})$ of real $n \times n$ traceless matrices. After that, Olshanetsky Perelomov and Kostant [10, 18, 19] used group-theoretic methods to integrate the equations of motion for the nonperiodic case. Kostant and Adler [1, 10] also showed that the phase space of the nonperiodic Toda lattice is symplectomorphic to a coadjoint orbit of the group of triangular matrices, as we will explain in detail further.

There are more recent works connecting Toda flows to numerical methods for computing eigenvalues of real symmetric matrices [6]. Also, it was shown in [11] that the set of points reached by flows of the Toda lattice through a real symmetric matrix with simple spectrum is diffeomorphic to a convex polytope, the *permutohedron*, using only linear algebra. In 2023, Leite Martínez Saldanha and Tomei [12] used charts of isospectral manifolds, introduced in [13], to transform the flows into linear differential systems with constant coefficients and diagonal matrices. In fact, the flows are converted into uniform straight line motion.

2.1 Lax representation, Flaschka's variables

The object of interest is the system of differential equations defined by the Hamiltonian (11). In other words, we have the following equations of motion

$$\dot{q}_k = \frac{\partial H}{\partial p_k} = p_k, \quad \dot{p}_k = -\frac{\partial H}{\partial q_k} = \exp(q_{k-1} - q_k) - \exp(q_k - q_{k+1}), \quad k = 1, \dots, n.$$

Clearly, the energy $H(q(t), p(t))$ is conserved. Notice that invariance under translation in configuration space (i.e. $H(q + c, p) = H(q, p)$ for any fixed constant $c \in \mathbb{R}^n$) implies uniform motion of the center of mass of the system: linear momentum is preserved. In other words, since the sum the derivatives of the p_k 's is zero, the center of mass of the system (arithmetic mean of q_k 's) moves with constant speed. Using Flaschka's variables [7],

$$a_k = -\frac{p_k}{2}, \quad k = 1, \dots, n \quad b_k = \frac{1}{2} \exp\left(\frac{q_k - q_{k+1}}{2}\right), \quad k = 1, \dots, n-1, \quad (12)$$

the evolutions become

$$\dot{a}_k = 2(b_k^2 - b_{k-1}^2), \quad \dot{b}_k = b_k(a_{k+1} - a_k). \quad (13)$$

Notice that if we know the initial position of the center of mass (mean of the initial positions), we know it for any given time, because the motion is uniform. Also, knowing the variables b_k is equivalent to knowing the relative positions $q_k - q_{k+1}$. And who knows the relative positions and their mean knows all positions (exercise)! This justifies the apparent loss of degrees of freedom on Flaschka's variables. Surprisingly, we can write equations (13) in matrix notation, obtaining a Lax pair

$$\dot{J} = [J, \Pi_{sk} J], \quad (14)$$

where J is a *Jacobi matrix*¹ whose diagonal and off-diagonal entries are the a_k and b_k respectively. The skew-symmetric projection of a matrix M , denoted by $\Pi_{sk}M$, is skew-symmetric and has the same lower triangular part as M . A relevant fact about Jacobi matrices is that their (real) spectrum is always simple. Also, first and last coordinates of eigenvectors are non-zero.

It can be shown that the eigenvalues of Jacobi matrices, together with the (normalized) first coordinates of eigenvectors constitute action-angle variables for the ordinary Toda lattice. In this case, the invariant Lagrangian submanifolds are non compact, since the isotropy subgroup of the action coming from the flows is trivial. In fact, the following result by Moser is a first step in the direction of this fact.

Theorem 2.1. *[8, 16] A Jacobi matrix is determined by its spectrum and the first coordinates of its appropriately normalized eigenvectors. Explicitly, set*

$$\mathcal{M} = \{(\lambda, c) = (\lambda_1, \dots, \lambda_n, c_1, \dots, c_n) \mid \lambda_1 < \dots < \lambda_n \text{ and } \sum_i c_i^2 = 1, c_i > 0\}$$

and $\psi : \mathcal{J} \rightarrow \mathcal{M}$, $\psi(J) = (\lambda_1, \dots, \lambda_n, c_1, \dots, c_n) = (\lambda, c)$, where λ_i and c_i denote the i -th smallest eigenvalue and the first coordinate of the associated normalized eigenvector of J , respectively. Then ψ is a diffeomorphism.

The hardest part of the proof is the construction of the inverse map, which we sketch. Let Λ be a diagonal matrix with entries given by λ and K be the invertible matrix with columns $c, \Lambda c, \dots, \Lambda^{n-1}c$. By a standard argument with Vandermonde determinants, K is invertible. The unique QR-factorization of an invertible matrix then gives

$$K = QR, \quad Q \in O(n), \quad R \in Up^+(n),$$

where $Up^+(n)$ denotes the group of $n \times n$ upper triangular matrices with strictly positive diagonal entries. Define $\tilde{J} = Q^T \Lambda Q$. The required matrix J is obtained by dropping the signs of the off-diagonal entries of $\tilde{J}$.

Remark 2.1. *The idea of Flaschka's variables was inspired on Peter Lax's work on casting the Korteweg-de Vries equation (KdV) into a... Lax pair form. In summary,*

$$u_t(t, x) = 6u(t, x)u_x(t, x) - u_{xxx}(t, x), \quad t \geq 0, x \in \mathbb{R} \quad (15)$$

is equivalent to the operator evolution $L'(t) = [L(t), H(t)]$, where L is known as the Strum-Liouville operator. The operators L and H act on functions as below:

$$\begin{aligned} L(t)f(x) &= -f''(x) + u(t, x)f(x) \\ H(t)f(x) &= 4f'''(x) - 6u(t, x)f'(x) - 3u_x(t, x)f(x). \end{aligned}$$

This implies that there are infinitely many conservation laws for this equation, which turns out to be enough for some notion of integrability...

It is easy to see that the Hamiltonian (11) becomes $H = \text{Tr}(J^2)/2$ in the new variables. Also, from the isospectral property of Lax pairs, we know that the eigenvalues of $J(t)$ do not change with t . Notice that a Lax pair does not always imply complete integrability: the conserved quantities may not be sufficient. Happily for us, the (always distinct) eigenvalues of J are indeed first integrals of the system, as we shall see next.

¹Jacobi matrices are real $n \times n$ symmetric tridiagonal matrices with strictly positive off-diagonal entries.

Proposition 2.1. *The solution of problem (14) with initial condition $J(0) = J_0$ is of the form $J(t) = Q^T(t)J_0Q(t)$, where $Q(t) \in SO(n)$ is such that $\dot{Q}(t) = Q(t)\Pi_{sk}J(t)$. In particular, $J(t)$ is defined for all $t \in \mathbb{R}$, i.e. the system has global solution.*

Proof. Simply take derivatives

$$\begin{aligned}\dot{J} &= -Q^T\dot{Q}Q^TJ_0Q + Q^TJ_0\dot{Q} \\ &= -Q^T\dot{Q}J + Q^TJ_0Q\dot{Q}^T \\ &= -Q^T\dot{Q}J + JQ^T\dot{Q} \\ &= [J, Q^T\dot{Q}] = [J, \Pi_{sk}J] .\end{aligned}$$

In particular, the matrix norm $\|J\|^2 = \text{Tr}(J^TJ)$ is conserved along the orbit. $\square$

Observe that symmetry and spectrum are preserved along the orbit $J(t)$. But what about Jacobiness? Why is this property also preserved? The answer lies on the next section. Consider the orbit starting from an arbitrary Jacobi matrix. We can get some information from the physical interpretation: informally, minimizing the potential energy (which is expected to happen as the system evolves) leads to the spreading of the particles, meaning that the entries b_k converge to zero as $t \rightarrow \infty$.

Proposition 2.2. *The limits of the orbits $J(t)$ for $t \rightarrow \pm\infty$ are diagonal matrices. The limit in the future (resp. past) corresponds to a diagonal matrix with strictly decreasing (resp. increasing) entries along the diagonal.*

Proof. The differential equations for the diagonal entries of $J(t)$ are given by $\dot{a}_k = 2(b_k^2 - b_{k-1}^2)$. Notice that the partial traces $\sum_{j=1}^k a_j$ are non decreasing. Since the norm of $J(t)$ is constant, all entries (and its derivatives) must remain bounded. This means that the rate of growth of the partial traces (which is essentially a sum of the non-diagonal entries) goes to zero. This implies the diagonal convergence of $J(t)$ at $\pm\infty$.

Now, notice that the sign of b_k is constant (and positive, by definition). Indeed, the differential equation $\dot{b}_k = b_k(a_{k+1} - a_k)$ admits solution constant equal to zero. But as $t \rightarrow \infty$, $b_k \rightarrow 0$, and thus $\dot{b}_k < 0$, implying the desired ordering of diagonal entries. Things work similarly as $t \rightarrow -\infty$. $\square$

As we first saw, the Toda lattice has a Hamiltonian formulation in $\mathbb{R}^{2n}$. After introducing Flaschka's variables and removing the (trivial) evolution of the center of mass, the phase space for the differential equation becomes the set of traceless Jacobi matrices, a cone of dimension $2n - 2$. Now, let the functions $\lambda_1, \dots, \lambda_n$ be the eigenvalues of J in the cone of traceless Jacobi matrices. Observe that since $a_k \approx p_k$ and $a_k \rightarrow \lambda_k$ as $t \rightarrow \pm\infty$, we have that $\{\lambda_i, \lambda_j\} \approx \{p_i, p_j\} = 0$, as $t \rightarrow \pm\infty$. But $\{\lambda_i, \lambda_j\}$ is constant along the Toda trajectories, by Poisson's theorem. This finally gives us the n desired first integrals needed to prove Liouville integrability.

2.2 Toda Flows in Coadjoint Orbits

Using Flaschka's variables, we can push forward the standard symplectic form to the cone of traceless Jacobi matrices, converting it into a symplectic manifold. Turns out that this symplectic structure also comes from coadjoint orbits, as we shall see.

Let $G = Up^+(n)$ be the group of $n \times n$ upper triangular real matrices with strictly positive diagonal entries (the connected component that contains the identity). Its Lie algebra $\mathfrak{g} = \mathfrak{U}$ is the vector space of real upper triangular matrices. We may identify $\mathfrak{g}^*$ with $\mathcal{S}$, the vector space of $n \times n$ real symmetric matrices via the non-degenerate bilinear form $\langle S, U \rangle = \text{Tr}(SU)$. For any $S \in \mathcal{S}$ and any $U \in \mathfrak{U}$, we have

$$\langle \text{Ad}_g^*(S), U \rangle = \langle S, \text{Ad}_{g^{-1}(U)} \rangle = \text{Tr}(Sg^{-1}Ug) = \text{Tr}(gSg^{-1}U) .$$

Observe that gSg^{-1} is not symmetric. However, its trace equals the trace of its projection to the symmetric component: consider the unique splitting

$$M = \Pi_{\mathcal{S}}M + \Pi_{\mathfrak{sU}}M , \quad (16)$$

where $\mathfrak{sU}$ denotes the vector space of real strictly upper triangular matrices. For any $E \in \mathfrak{sU}$ and $U \in \mathfrak{U}$ we have $\text{Tr}(EU) = 0$. Thus, $\langle \text{Ad}_g^*(S), U \rangle = \text{Tr}(gSg^{-1}U) = \text{Tr}(\Pi_{\mathcal{S}}(gSg^{-1})U)$, and so we finally obtain

$$\text{Ad}_g^*(S) = \Pi_{\mathcal{S}}(gSg^{-1}) .$$

Now, start with

$$S = \begin{pmatrix} 0 & 1 & & \\ 1 & 0 & \ddots & \\ & \ddots & \ddots & 1 \\ & & 1 & 0 \end{pmatrix} .$$

The coadjoint orbit passing through S is $\mathcal{O}_S = \{\Pi_{\mathcal{S}}(gSg^{-1}) | g \in \mathcal{U}^+\}$ and it consists of traceless Jacobi matrices. The symplectic structure on $\mathcal{O}_S$ is given by the Kirillov-Kostant-Souriau (KKS) form:

For every $J \in \mathcal{O}_S$, the tangent space $T_J\mathcal{O}_S$ is spanned by the infinitesimal generators of the coadjoint action, which are given by

$$U_S(J) = \left. \frac{d}{dt} \right|_{t=0} \text{Ad}_{\exp(tU)}^*(J) = \Pi_{\mathcal{S}}[U, J] .$$

Thus, the KKS form behaves like

$$\omega_J(U_S(J), V_S(J)) = -\langle J, [U, V] \rangle = -\text{Tr}(J[U, V]) .$$

We finish with the following proposition.

Proposition 2.3. *The Toda lattice is the vector field X_H associated to the Hamiltonian $H(S) = \text{Tr}(S^2)/2$ defined on the coadjoint orbit $\mathcal{O}_S$.*

Proof. Let U and $V \in \mathfrak{U}$ be associated to X_H and $Y \in \mathfrak{X}(\mathcal{O}_S)$, respectively (so in particular, $Y(J) = V_S(J) = \Pi_{\mathcal{S}}[U, J]$). Then,

$$\omega_J(X_H, Y) = \omega_J(U_S(J), V_S(J)) = -\text{Tr}(J[U, V]) = -\text{Tr}([J, U]V) .$$

On the other hand, we have

$$\omega_J(X_H, Y) = d_J H(Y) = \text{Tr}(JY(J)) = \text{Tr}(J \Pi_{\mathcal{S}}[V, J]) .$$

Let $\mathcal{A}$ denote the vector space of $n \times n$ skew-symmetric matrices. Consider another unique splitting $M = \Pi_{\mathfrak{U}}M + \Pi_{\mathfrak{sU}}M$, where $\Pi_{\mathfrak{sU}}M \in \mathcal{A}$ has the same lower part as M , as

in (14). Making use of the orthogonalities (with respect to the trace form) $\mathcal{A} \perp \mathcal{S}$ and $\mathfrak{U} \perp \mathfrak{sU}$, we obtain

$$\begin{aligned} -Tr([J, U]V) &= Tr(J \Pi_{\mathcal{S}}[V, J]) \\ &= Tr(\Pi_{\mathfrak{U}}J \Pi_{\mathcal{S}}[V, J]) \\ &= Tr(\Pi_{\mathfrak{U}}J[V, J]) \\ &= Tr([J, \Pi_{\mathfrak{U}}J]V) \\ &= -Tr([J, \Pi_{sk}J]V) . \end{aligned}$$

Thus, $Tr([J, U]V) = Tr([J, \Pi_{sk}J]V)$ for every $V \in \mathfrak{U}$. Therefore, $[J, U] = [J, \Pi_{sk}J] + E$ for some $E \in \mathfrak{sU}$. Since $[\mathcal{S}, \mathcal{A}] \subset \mathcal{S}$, we have

$$\dot{J} = X_H(J) = \Pi_{\mathcal{S}}[J, U] = \Pi_{\mathcal{S}}[J, \Pi_{sk}J] = [J, \Pi_{sk}J] ,$$

as desired. □

This leads to the geometric interpretation of the fact that the solution of (14) is always Jacobi: it simply does not leave the coadjoint orbit.

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